Name_____ Date_____ Period_____

LESSON 5-8 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Surface Area of Pyramids

Maryland Standard 6.G.A.4 · Learning target – I can find the surface area of a pyramid by adding the base area and the areas of its triangular faces.

REMEMBER

- SA of a pyramid = **base area** + area of all the **triangular faces**.
- Each triangle's height is the **slant height**, not the height of the pyramid.

WORKED EXAMPLE

Square pyramid: base 8 cm, slant height 7 cm.

Step 1 Base: 8 × 8 = 64

Step 2 One triangle: ½ × 8 × 7 = 28; four: 112

Step 3 Add: 64 + 112 = 176

Answer **176 cm²**

WORD BANK

slant height – height of a side triangle, along its face

base – the bottom face

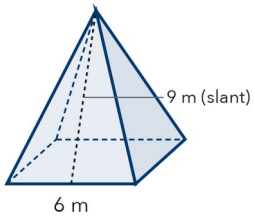
triangular face – a side triangle

surface area – base + all the triangles

PRACTICE

show your work · label units

1. Find the surface area of the square pyramid.



Step 1 Base: 6 × 6.

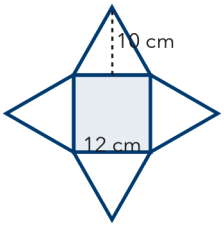
Step 2 One triangle: ½ × 6 × 9.

Step 3 Add the base and all 4 triangles.

Show your work.

Answer_____ m²

2. Use the net of the square pyramid.



Step 1 Base: 12 × 12.

Step 2 Triangle: ½ × 12 × 10.

a. Area of the base _____ cm²

b. Area of one triangle _____ cm²

Surface area_____ cm²

3. Find the surface area of each square pyramid.

a. base 4 in, slant height 5 in _____ in²

b. base 10 cm, slant height 12 cm _____ cm²

4. Gift Box. A gift box is a **square pyramid**. The base is 9 in on each side. The slant height is 11 in. How much paper covers **every** face?

Step 1 Find the base.

Step 2 Find one triangle, then four.

Step 3 Add.

Show your work.

Answer_____ in²

5. Find the Error. A square pyramid has a base 6 ft on each side, a height of 4 ft, and a slant height of 5 ft. Kofi wrote SA = 36 + 4 × (½ × 6 × 4) = 84 ft². Explain his mistake and find the surface area.

Frame *Kofi used the ___ instead of the ___. The surface area is ___.*

_____

_____

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Use a folded paper tent to show someone a pyramid's height and its slant height.

Name_____ Date_____ Period_____

LESSON 5-8 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Lateral Area of Pyramids

Maryland Standard 6.G.A.4 · Learning target – I can find the lateral area of a pyramid, which covers only its slanted faces.

REMEMBER

- **Lateral area** is the area of the slanted faces only. Do **not** include the base.
- Lateral area = surface area − base area.

WORKED EXAMPLE

Square pyramid: base 10 m, slant height 13 m. Find the lateral area.

Step 1 One face: ½ × 10 × 13 = 65

Step 2 Four faces: 4 × 65 = 260

Answer **260 m²**

WORD BANK

lateral area – only the slanted faces

lateral face – a slanted side triangle

no floor – leave out the base

bundle – one pack of shingles

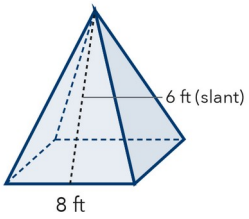
PRACTICE

show your work · label units

1. Find the lateral area of the square pyramid.

Step 1 One triangle: ½ × 8 × 6.

Step 2 Multiply by 4. Do not add the base.



Show your work.

Answer_____ ft²

2. Find each lateral area.

a. SA = 200 in², base = 64 in² _____ in²

b. 4 faces, each 18 cm² _____ cm²

c. square base, side 6 m; slant height 10 m _____ m²

d. square base, side 3 ft; slant height 4 ft _____ ft²

3. **Tower Roof.** A roof is a **square pyramid**: base 14 ft on each side, slant height 10 ft. Shingles cover **only** the slanted faces. One bundle covers 35 ft². How many bundles?

Step 1 Find the lateral area.

Step 2 Divide by 35.

Answer_____ bundles

4. Lampshade. A lampshade is a square pyramid with no base. Its lateral area is 160 in² and each base edge is 8 in. What is its slant height?

Step 1 One face: 160 ÷ 4.

Step 2 ½ × 8 × slant height = one face. Find the slant height.

Answer_____ in

5. Play Tent. A play tent is a square pyramid with a base 5 ft on each side and a slant height of 6 ft. It has no floor. Sasha says it needs 85 ft² of fabric. Is she right? Explain.

Frame *Sasha is ___ because the tent has no ___, so ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. A square pyramid has a base 4 cm on each side and a slant height of 7 cm. What is its lateral area?

A 14 cm² **B** 56 cm² **C** 72 cm² **D** 112 cm²

7. A pyramid has a surface area of 120 m² and a lateral area of 84 m². What is its base area?

A 204 m² **B** 120 m² **C** 36 m² **D** 42 m²

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___